

CASTLEPOINT, New Zealand

WMO: 934980

Lat: 40.9042S

Lon: 176.2119E

Elev: 120

StdP: 99.89

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	5.3	-0.6	3.7	6.6	0.5	4.0	6.9	31.3	10.0	26.6	11.5	7.6	240	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.4	24.6	17.8	23.1	17.1	21.9	16.6	18.9	22.6	18.2	21.3	17.5	20.4	10.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.9	20.1	17.0	12.3	19.2	16.3	11.8	18.8	54.3	22.7	51.8	21.5	49.7	20.5	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	22.1	19.5	DB	3.0	27.7	0.7	1.8	2.5	28.9	2.0	30.0	1.6	31.0	1.1	32.3	(4)
(5)			WB	1.5	20.7	0.7	1.0	0.9	21.4	0.5	21.9	0.1	22.5	-0.4	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	17.3	17.5	16.4	14.3	12.6	10.5	9.7	9.8	11.1	12.4	13.6	15.7	(6)
(7)		DBStd	3.63	2.71	2.59	2.45	2.35	2.50	2.20	2.03	2.05	2.34	2.37	2.59	2.70	(7)
(8)		HDD10.0	108	0	0	0	1	7	19	30	28	14	7	2	0	(8)
(9)		HDD18.3	1864	53	42	69	123	179	234	267	263	216	185	143	91	(9)
(10)		CDD10.0	1346	226	211	199	129	86	36	22	22	49	80	111	176	(10)
(11)		CDD18.3	60	20	20	10	1	0	0	0	0	0	0	1	8	(11)
(12)		CDH23.3	116	44	48	11	0	0	0	0	0	0	0	1	12	(12)
(13)		CDH26.7	7	3	4	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.6	8.8	8.0	7.9	7.5	8.2	8.3	8.7	8.4	9.3	9.8	9.6	8.8	(14)
(15)	Precipitation	PrecAvg	1087	69	75	78	82	96	114	142	109	81	97	74	66	(15)
(16)		PrecMax	1526	246	279	244	294	221	233	285	230	207	214	170	149	(16)
(17)		PrecMin	615	11	19	19	12	13	55	54	44	26	46	18	14	(17)
(18)		PrecStd	218	50	61	49	69	46	40	60	40	40	42	37	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.7	27.2	24.7	21.6	19.4	16.8	15.8	16.0	18.2	19.8	22.4	24.8	(19)
(20)			MCWB	18.4	19.3	16.9	16.3	15.3	13.4	12.4	12.1	13.4	14.3	16.0	17.5	(20)
(21)		2%	DB	24.5	24.8	22.7	19.8	18.2	15.6	14.4	14.6	16.8	18.2	20.2	22.9	(21)
(22)			MCWB	17.7	18.2	16.7	15.0	14.4	12.2	11.2	10.9	12.6	13.5	14.9	16.7	(22)
(23)		5%	DB	22.7	23.0	21.2	18.7	17.0	14.6	13.4	13.6	15.6	17.2	18.8	21.3	(23)
(24)			MCWB	16.8	17.5	16.1	14.6	13.3	11.6	10.4	10.2	11.9	13.0	14.3	16.1	(24)
(25)		10%	DB	21.3	21.5	19.9	17.6	16.0	13.8	12.6	12.7	14.6	16.1	17.6	19.8	(25)
(26)			MCWB	16.4	16.9	15.7	14.2	12.8	11.1	9.9	9.8	11.2	12.3	13.5	15.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.1	20.4	18.5	17.5	16.1	14.2	13.0	12.7	14.4	15.3	17.0	19.0	(27)
(28)			MCDWB	23.8	24.5	21.7	19.5	18.3	15.8	14.9	14.8	16.9	18.5	20.6	22.7	(28)
(29)		2%	WB	18.9	19.1	17.6	16.3	15.1	13.0	11.8	11.7	13.2	14.2	15.8	17.7	(29)
(30)			MCDWB	22.3	23.1	20.6	18.3	17.0	14.5	13.5	13.4	15.9	17.2	19.1	20.9	(30)
(31)		5%	WB	18.1	18.3	17.0	15.6	14.3	12.2	11.0	10.9	12.4	13.5	14.9	16.9	(31)
(32)			MCDWB	21.0	21.7	19.6	17.5	16.1	14.0	12.9	12.7	14.8	16.3	17.6	19.8	(32)
(33)		10%	WB	17.3	17.5	16.4	14.9	13.5	11.5	10.3	10.3	11.6	12.8	14.1	16.1	(33)
(34)			MCDWB	20.2	20.5	19.0	16.9	15.5	13.5	12.1	12.2	13.9	15.4	16.7	18.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	5.4	5.0	4.5	4.1	3.9	3.6	3.9	4.5	5.2	5.3	5.3	(35)
(36)			MCDBR	7.9	7.9	6.9	5.9	4.9	4.7	4.2	5.0	5.7	6.5	7.1	7.4	(36)
(37)			MCWBR	4.3	3.8	3.6	3.5	3.3	3.5	3.1	3.5	3.9	4.2	4.4	4.1	(37)
(38)		5% WB	MCDBR	6.5	6.5	5.4	4.4	4.0	3.6	4.0	3.6	5.0	5.8	6.1	6.2	(38)
(39)			MCWBR	4.0	4.0	3.4	3.4	3.3	3.4	3.1	3.3	3.8	4.2	4.2	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.331	0.322	0.307	0.290	0.294	0.289	0.298	0.317	0.328	0.322	0.334	(40)	
(41)		taud	2.558	2.597	2.609	2.645	2.670	2.618	2.640	2.627	2.549	2.532	2.575	2.579	(41)	
(42)		Ebn at Noon	990	972	939	889	843	798	832	886	928	965	1001	1000	(42)	
(43)		Edh at Noon	105	97	87	73	60	58	61	72	91	102	103	104	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.59	5.72	4.44	3.03	2.05	1.52	1.75	2.55	3.75	5.04	6.10	6.42	(44)	
(45)		RadStd	0.39	0.52	0.29	0.23	0.13	0.10	0.14	0.17	0.26	0.30	0.47	0.45	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		N/A	+0.32	+0.50	N/A	N/A	N/A	-25	N/A	N/A	N/A	

Nomenclature: See separate page